

LeetCode Shortcuts & Problem-Solving Patterns

1. Quick Pattern Recognition

Problem clue	Think of
Find duplicate	HashSet
Count frequency	HashMap / array
Two numbers → target	Two Sum / HashMap
Sorted array + pair	Two Pointers
Subarray / substring	Sliding Window
Maximum/minimum subarray	Kadane's Algorithm
Sorted array search	Binary Search
Reverse linked list	3 pointers
Tree level-by-level	BFS / Queue
Tree depth/path	DFS / Recursion
Shortest path in unweighted graph	BFS
Connected components	DFS / BFS / Union-Find
Matching brackets	Stack
Next greater element	Monotonic Stack
Top K elements	Heap / PriorityQueue
All combinations/permutations	Backtracking
Choose / not choose	DP
Grid movement	DFS / BFS
Prefix-based strings	Trie

2. Java Shortcuts

```
Arrays.sort(arr);  
  
Math.max(a, b);  
Math.min(a, b);  
  
HashSet<Integer> set = new HashSet<>();  
HashMap<Integer, Integer> map = new HashMap<>();  
map.put(x, map.getOrDefault(x, 0) + 1);  
Deque<Integer> stack = new ArrayDeque<>();  
stack.push(x);  
stack.pop();  
stack.peek();  
  
Queue<Integer> q = new LinkedList<>();  
q.offer(x);  
q.poll();
```

```
PriorityQueue<Integer> pq = new PriorityQueue<>();  
PriorityQueue<Integer> pq = new PriorityQueue<>(Collections.reverseOrder());
```

3. Recommended Learning Order

Array → HashMap → Two Pointer → Sliding Window → Binary Search → Stack → BFS/DFS → DP

Tip: For very easy LeetCode problems, focus first on recognizing the pattern before writing code.